

A 97% Peak Efficiency 12V to 1V 500A Multi-Phase Direct Power Converter for High Current Applications

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Abstract—This paper presents a 12-1V fixed gain Multi-Phase Direct Power Converter, a one-stage converter for high current applications. Based on the Direct Power Converter and its operation principles, it enables duty cycles up to 85 to 90% for all the power switches, reducing drastically the rms current in the power devices, zero voltage switching (ZVS) in both primary and secondary side and zero current switching (ZCS) in secondary side. Due to the expected high output current, Segmented Winding Transformer (SWT) was also implemented to reduce secondary side losses. A prototype with four phases parallel connected is designed and build to split the high output current into multiple parallel current paths, allowing to reduce the output voltage ripple and the overall capacitance as the phases can be shifted. The prototype was tested up to 550A, achieving a peak efficiency of 97% at 200A and 95% efficiency at full load (500A) with a switching frequency of 150kHz, which allows to parallelize power switches without increasing driving losses dramatically.

Keywords— Direct to Chip, Power Converters, High-Current, DCx, Vertical Power, segmented winding transformer, synchronous rectification, dc-dc

I. INTRODUCTION

In the last years, the trend has shown the increasing size and consumption of microprocessors (CPUs, GPUs, TPUs, etc.), which has led to a great advance in power electronics, both at component package level (improving the parameters of the devices) and at physical implementation of the converters, leading to 3D implementations ([1-2]). Due to the extremely high currents required in new applications, power converters with very high efficiency, high power density and high current density are required to supply the actual and future microprocessors. Many different topologies and implementations have been developed and proposed, as multi-phase buck designs or two-stage hybrid switched-capacitors designs [3], [4], [5]. Among all the solutions “Direct Power Converter -DPx-” ([6-7]) emerges as a very interesting solution. This topology allows all the power devices to switch simultaneously with duty cycles up to 90% (only limited by the breakdown voltage of the devices), which is a key advantage for very high output current applications as it permits to reduce the losses dramatically. This topology in combination with Segmented Winding Transformer (SWT) [8] seems to be very promising solution for the present and future high current low voltage applications.

This paper presents a step forward with the design and implementation of a Multi-Phase Direct Power Converter and shows enhanced experimental results with a 12V to 1V prototype which achieves a peak efficiency of 97%, full load efficiency of 95% and a power density of 279 W/in³.

II. POWER TOPOLOGY & OPERATION

A. Power topology

Direct Power Converter -DPx- appear like a Flyback converter but with the corresponding terminals of the transformer of a Forward converter ([9-11]) (see Fig.1). DPx converter is conceived as a DCx with insignificant series inductance, which allows the current to be flat during the on time. The conduction time achieves values up to 90% (only limited by the breakdown voltage of the devices) thus reducing the current rms values in both primary and secondary sides. The reset of the transformer is produced by the resonance between the magnetizing inductance and parasitic (and added) capacitances of the primary switches. Besides, this power topology allows some advantages as Zero Voltage Switching (ZVS) in both primary and secondary switches, Zero Current Switching (ZCS) or no load operation.

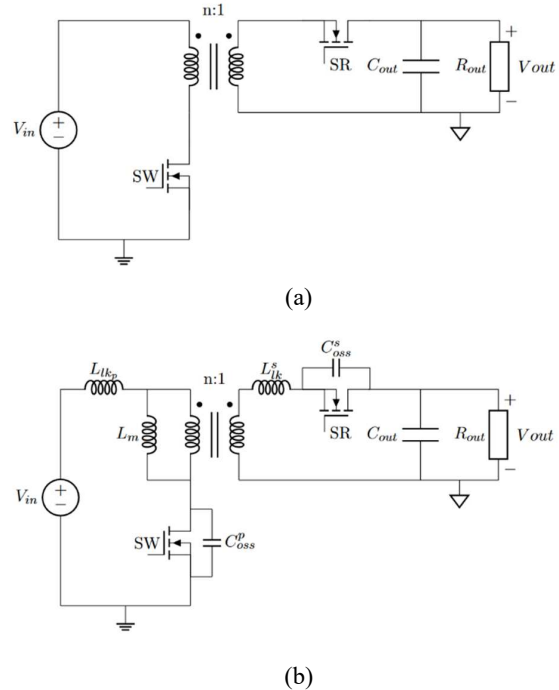


Fig. 1. Direct Power Converter Topology. (a) Power topology considering ideal transformer. (b) Power converter topology considering parasitic elements of the transformer: both primary and secondary leakage inductance.

B. Circuit operation

The key of the Single-Phase Direct Power Converter is that all the switches are in ON state and OFF state simultaneously, enabling duty higher than 50% (up to 90%)

and hence reducing the rms current. A direct power path between the input and the output of the converter is created through the transformer in ON state of both switches. The current flowing through primary and secondary can reach steady state if leakage inductance is small enough. The level of the current is roughly I_o/d , being “d” the duty cycle. More relevant waveforms of the single-phase DPx are shown in Fig. 2.

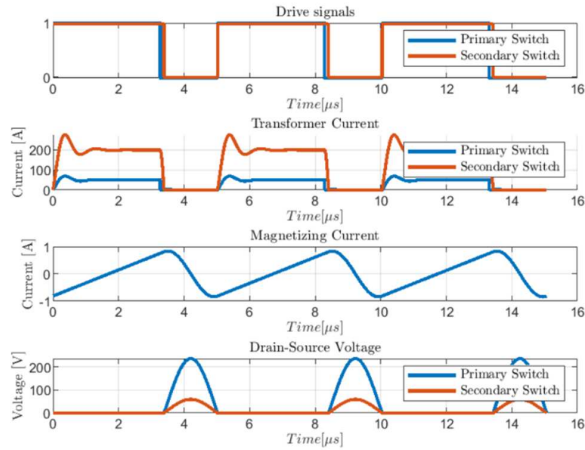


Fig. 2. Key waveforms of the Direct Power Converter. From top to bottom: driving signals of primary and secondary switches, primary and secondary transformer current, magnetizing current and drain to source voltage in primary and secondary switches.

Each cycle has three different states. During the first state, both primary and secondary switches are conducting (ON) simultaneously, creating a direct path between the input and output of the converter through the transformer. In this state, the power flows directly from the input to the output. Magnetizing current raises during ON time, starting from a negative value which is approximately equal to the positive value and the end of the state (see Fig. 3).

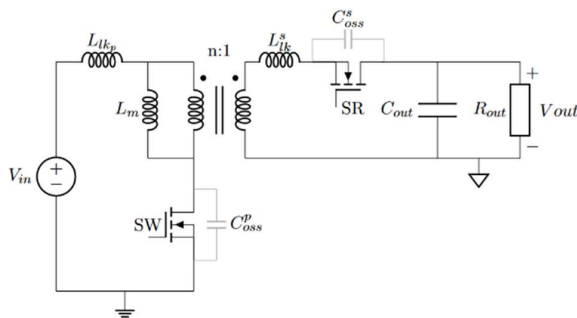


Fig. 3. First state of the cycle. Both primary and secondary switches are conducting simultaneously.

In the second state, primary switch is opened (OFF) while maintaining secondary switch in ON state. Thus, the current drops to zero and we can have Zero Current Switching (ZCS) in secondary switches (see Fig. 4). In the third state, we keep primary switch opened and secondary switch is opened as well. The magnetizing inductance resonates with the output capacitance of the primary switches (plus added capacitance if required) thus producing the resonance between C_{oss} and L_m and the reset of the transformer (see Fig. 5).

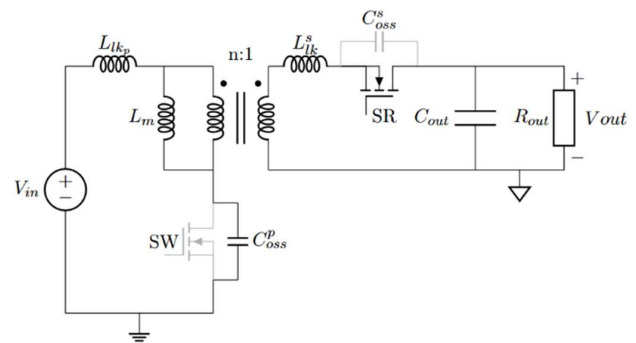


Fig. 4. Second state of the cycle. Primary switch is opened while maintaining secondary switch closed.

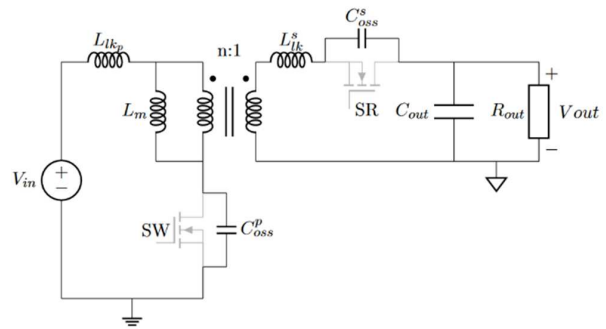


Fig. 5. Third state of the cycle. Both primary and secondary switches are opened and the reset of the transformer is produced.

In the Multi-Phase Direct Power Converter (see Fig. 7), each phase can be shifted freely which leads to some benefits that are very suitable for integrated systems as microprocessors or in applications where high current, high current steps and small output ripple is required. Some of the advantages of multi-phase converters are:

- Reduced input capacitance
- Reduced output capacitance
- Improved thermal performance and efficiency at high load currents
- Improved overshoot and undershoot during load transients

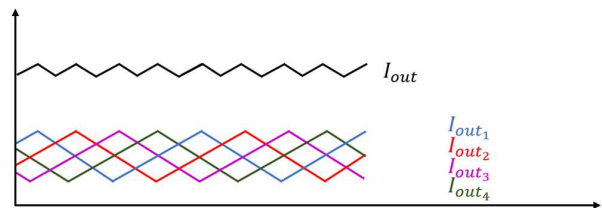


Fig. 6. Example of current ripple cancellation in multi-phase converters.

The interleaved nature of the phase timing in a multiphase converter reduces the maximum and RMS value of the current drawn by the switch input, thus the input capacitance can be reduced while maintaining equivalent ripple performance. Similarly occurs with the output side of the converter, each phase of the converter supply part of the required output current, being the output ripple consistent from phase to phase.

However, the sum of the currents has a lower ripple as the current variations from each phase do not occur at the same time (see Fig. 6).

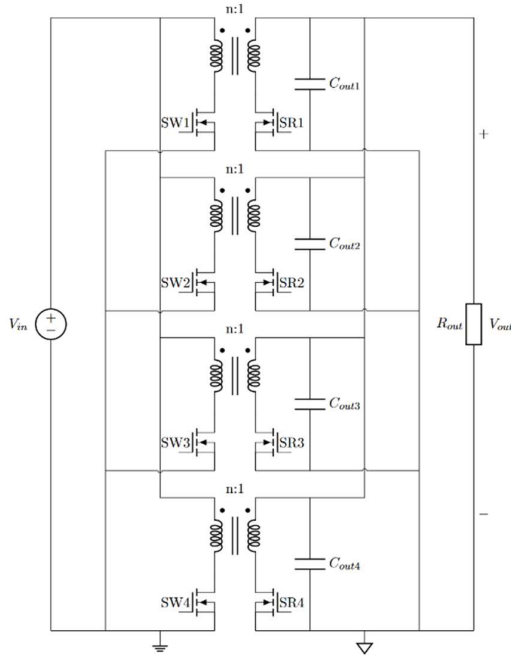


Fig. 7. Multi-Phase Direct Power converter structure.

Normally, in very high current applications conduction losses are dominant. Typically, conduction losses are reduced by decreasing the RMS value of the current through the power switches enabling higher duty cycle. This can't be done easily in structures in many converter topologies (multi-phase buck converters) because enabling higher duty cycles (over 50%) for one of the ports implies using lower duty cycle for the other port (lower than 50%). In the case of the Direct Power Converter, all the switches conduct simultaneously rather than alternately, allowing to increase the duty cycle up to 90% (only limited by the breakdown voltage of the devices).

Applying the volt.second balance equal to zero to the voltage waveform in the transformer (see Fig. 8), the maximum voltage applied to the power switch can be calculated as:

$$V_{max} \approx \frac{\pi}{2} \frac{V_{in}}{1-D} \quad (1)$$



Fig. 8. Voltage waveform in the power transformer of the converter.

This maximum voltage is used in combination with Fig. 9 to choose the optimum power switch for both primary and secondary switches. This picture help to choose the optimum duty cycle as we relate the breakdown voltage of the devices, the specific device resistant and the specific output resistant considering only conduction losses for high current applications.

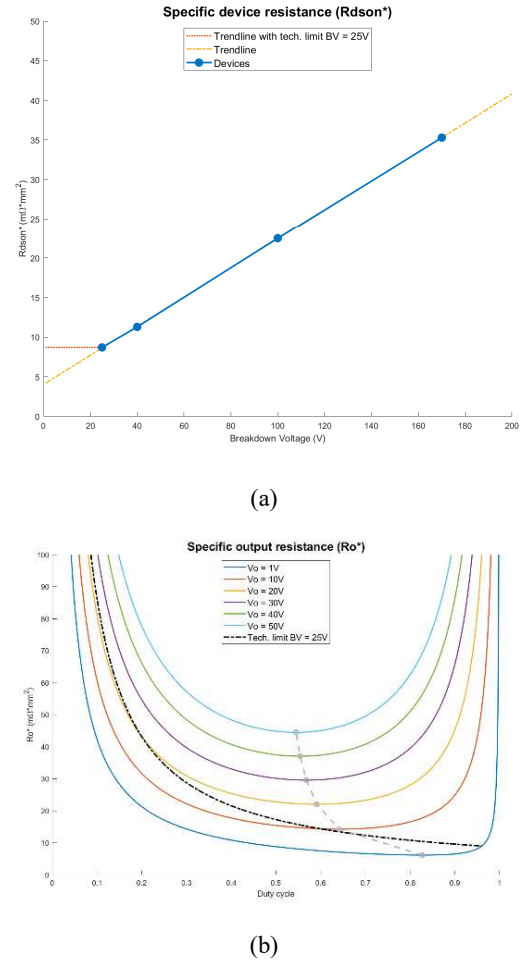


Fig. 9. (a) Specific device resistance depeneding on breakdown voltage. (b) Specific output resistance of a converter depending on duty cycle and output voltage considering only conduction losses.

In very high current applications, conduction losses become dominant and the most important losses to consider in the losses calculation for a converter. The efficiency of a converter can be calculated as follows if only conduction losses are considered:

$$\eta = \frac{V_o * I_{o\Box}}{V_o * I_{o\Box} + I_{o\Box}^2 * R_o^*} \quad (2)$$

$$\eta = \frac{V_o}{V_o + I_{o\Box} * R_o^*} \quad (3)$$

From equation (3) the equivalent output resistance of a converter can be obtained as:

$$R_o^* = \left(\frac{1}{\eta} - 1\right) \frac{V_o}{I_{o\Box}} \quad (4)$$

$$R_o^* = \frac{R_o * A_T}{I_o^2} \sum_{j=1}^n \frac{I_{rmsj}^2}{A_j} \quad (5)$$

From this equation it can be seen that the output resistance of the converter depends directly on the specific device resistance (which depends on both the breakdown voltage of the device and the technology) and with the duty cycles of the switches as the higher the duty cycle , the lower the rms current value respecting the mean value of the current.

III. MAGNETIC STRUCTURE

One of the fundamentals of the converter is the implementation of a Segmented Winding Transformer (SWT) for the output stage. The output capacitor and the Synchronous Rectifier (SR) are placed in the opposite sides of the magnetic core (see Fig. 10). Both, the capacitor and the switch replace part of the secondary winding, thus reducing to the minimum the output current path, which makes it ideal for high current applications. Besides, many devices can be paralleled efficiently to reduce conduction losses.

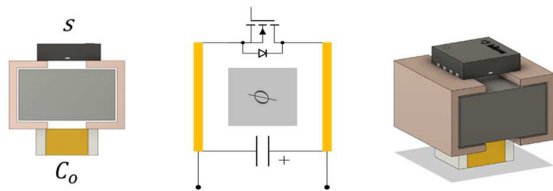
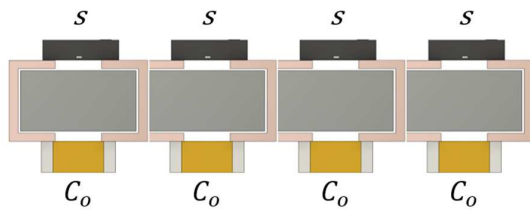
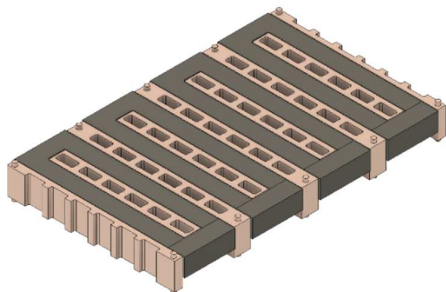


Fig. 10. Segmented Winding Transformer (SWT): Output capacitor and power switch are placed on opposite sides of the magnetic core.

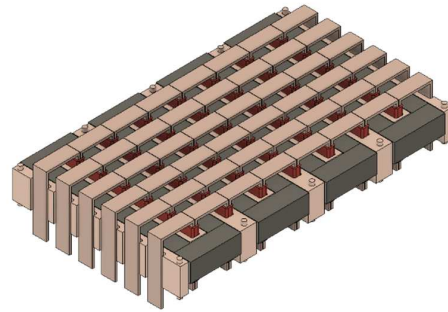
For the case of the multi-phase implementation, the structure is replicated in each module. In each phase, two external bus bars and one center bus bar are placed around the magnetic core. Those external bus bars are shared by some of the phases. The secondary loop is closed by the synchronous rectifiers and the output capacitors (see Fig. 11). To reduce the ac resistance of the windings and the leakage inductance, primary windings are embedded into secondary bus bars (see Fig. 11). High frequency effects in the windings are minimized (both proximity effect and skin effect). Besides, due to the high mean value of the current (due to a high duty cycle), the harmonic content is low.



(a)



(b)



(c)

Fig. 11. Magnetic structure configuration. (a) Front view of the secondary side magnetic structure. (b) Secondary windings bus bars location. (c) Primary windings embedded inside secondary windings to reduce leakage inductance.

IV. EXPERIMENTAL RESULTS

To verify the effectiveness of the Multi-Phase Direct Power Converter, a prototype was designed and tested (see Fig. 12) mainly for 12V to 1V but also for an output voltage of 0.5V, 0.75V and 1.2V. The area of the power stage is 52.5mm x 64mm and the height of the converter is 8.2mm. The transformer resembles a toroidal transformer despite the core is square (see Fig. 11). There are two external bus bars and a central one for each magnetic core to carry the output current. The prototype consists of three different PCBs (see Fig. 12) whose main elements are shown in Table I and Table II.

TABLE I. MAIN COMPONENTS OF THE PROTOTYPE

Specification	Device	Description
Primary Side Switch	EPC2215	200V 8mΩ
Secondary Side Switch	IQE006NE2LM5CGATMA1	25V 0.65mΩ
Primary Side Driver	UCC27611	5V 4A/6A
Secondary Side Driver	2EDN7524	5V 5A
Input Capacitors	C1206C474MMRECLR	470nF 63V 1206
Output Capacitors	GRM31CR60J227ME1L	220uF 6.3V 1206

TABLE II. TRANSFORMER SPECIFICATIONS

Element	Value
Core Type	E-I Custom Core (4.2mm height)
Core Material	3C95
Primary Winding	12turns 400um x 3.5mm
Secondary Winding	1 Turn 3.9mm x 6.4mm

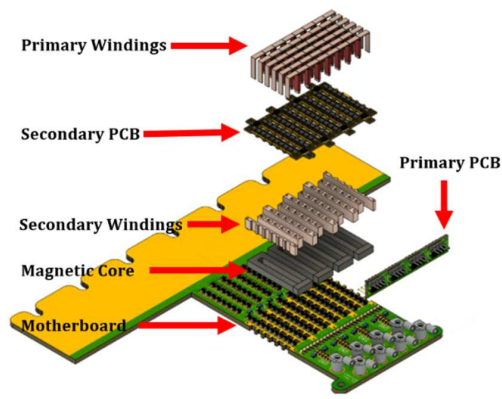


Fig. 12. Prototype distribution. From top to bottom and from left to the right: primary windings, secondary PCB, secondary windings, magnetic core, primary PCB and motherboard. The dimensions of the converter are 52.5mm x 64mm x 8,2mm.

In the primary side, 200V GaN devices are used (4x EPC2215 each module). These switches are needed to achieve high duty cycle, which is one of the key parameters of the topology. An additional CDS capacitance is added to adjust the resonance between the inductance and the parasitic output capacitance of the device.

In the secondary side 20 synchronous rectifiers (IQE006NE2LM5CG) are paralleled in each module to have low conduction resistance. This leads to a high filling factor of synchronous rectifiers, referring to the proportion of area which is occupied by the power switches in the secondary side PCB (see Fig. 13). The total area of the PCB is 2656mm² and the total area occupied by the synchronous rectifiers is 1155 mm², which means a filling factor of 43.49%.

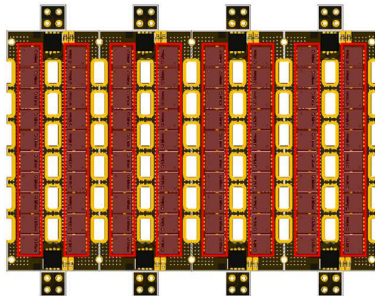


Fig. 13. Area distribution of the Synchronous Rectifiers in the secondary PCB.

The converter can operate at no load conditions (see Fig. 14) and the optimal point of operation shows with the highest possible duty cycle, which is only limited by the voltage apply in the switch by the resonance to reset the transformer.

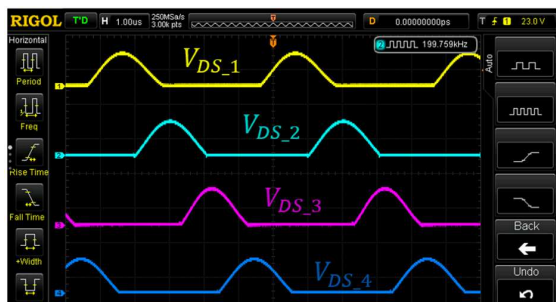


Fig. 14. Voltage applied to each module of the converter with all the phases shifted.



Fig. 15. Voltage applied to the primary switching devices at no load operation for 12V input voltage. Yellow: module 1. Cyan: module 2. Magenta: module 3. Blue: module 4.

First, the converter is tested for an output voltage of 1V to supply a maximum output current of 500A (see Fig. 16). The converter operates at low frequency (around 150kHz) and with duty cycles up to 88%. Peak efficiency is 97% at 200A (6.3W total loss) and full load efficiency of 95% (26.27W total loss).

This prototype was also tested to generate 0.5V, 0.75V and 1.2V. For the case of 0.5V, a peak efficiency of 97.4% and full load (500A) of 91.03%. For an output voltage of 0.75V, peak efficiency of 97.13% and 93.73% full load efficiency (500A) is obtained. Finally, for an output voltage of 1.2V, 97.03% peak efficiency and full load (450A) efficiency of 96% is achieved (see Fig. 17).

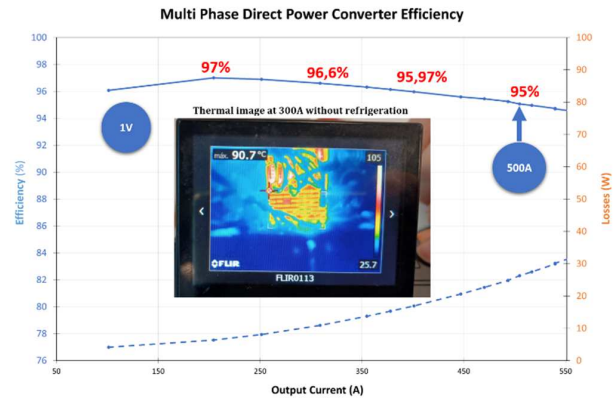


Fig. 16. Measured converter efficiency and thermal image of the prototype at 300A without refrigeration.

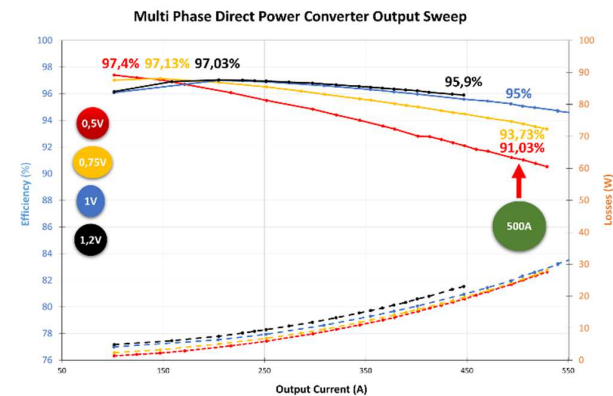


Fig. 17. Measured converter efficiency of the prototype for an output voltage from 0.5V to 1.2V.

V. CONCLUSIONS

This paper presents and validates a Multi – Phase Direct Power Converter with a fixed gain of 12, which may be higher by changing the number of turns, for high output current applications. It features Zero Voltage Switching (ZVS) in both primary and secondary switches and Zero Current Switching (ZCS) in secondary side. Moreover, by employing Segmented Winding Transformer (SWT), the proposed topology can achieve high efficiency in high current applications. With a footprint of 52.5mm x 64mm it has been able to supply up to 550A. The input voltage is converted into 0.5-1.2V.

For an output voltage of 1V, the efficiency is over 94.6% for the whole range of operation, with a peak efficiency of 97% at 200A and 95% efficiency at 500A, which highlights the flatness of the efficiency curve.

The prototype was also tested for and output voltage of 0.5V, 0.75V and 1.2V. Obtaining some key points as 95.9% at 450A for 1.2V or 97.4% peak efficiency for 0.5V.

Overall, all these results are very good and promising, showing the suitability of this topology for low voltage and high current applications.

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